

Work Experience

Mechanical and Materials Engineering Co-op | Liburdi GAPCO | January 2026 – April 2026

- Authored 7 weld maps as the designated draftsman detailing heating protocol, joint design, dimensional requirements, and bead placement for nuclear welding applications
- Analyzed dilution and ferrite content of sample welds for parameter adjustment and quality control
- Fixed gaps in existing documentation and progressed new projects by preparing preheat, thermocouple attachment, weld repair, and welder instruction procedures

Undergraduate Research Assistantship (URA) | University of Waterloo | September 2025 – December 2025

- Prototyped a vibration damping apparatus to test actuators for precision machinery research

Manufacturing Engineering Co-op | Martinrea Tillsonburg | May 2025 – August 2025

- Designed, fabricated, and installed a sheet metal part ejection system in 9 sub-cells resolving a critical issue which previously caused over 2.5 hours of downtime
- Updated tooling large subassemblies and relocated spot welds to reduce cycle time by over 4.5 seconds
- Designed two placement verification systems ensuring proper alignment, saving over \$580 in lost parts

Library Assistant | Uxbridge Public Library | September 2023 – August 2024

Skills

SolidWorks (CSWA); SolidEdge – Strong skills in 3D modelling, complex assembly design, and technical drafting

GMAW/GTAW Welding – Welding fundamentals and safety. Crafted a scorpion out of nuts, bolts, and washers

PLC and FPGA – Basic ladder logic and VHDL programming

Programming Languages – C++, Python, PHP, MySQL, HTML, CSS

Projects

Stratospheric Balloon | April 2025 – present

- Mechanical lead of a self-started design team creating a controllable data collecting balloon with proposed altitude of 30km set to launch in August 2026
- Responsible for design and execution of the payload casing, valve system, termination system, and protection of electronics

Autonomous Pipe Cleaning Robot | October 2024 – November 2024

- Mechanical designer of a team that fabricated a pipe cleaning robot for the MTE 100 class.

Education

Mechatronics Engineering (Co-op), Bachelor of Applied Science - BASc, University of Waterloo | 2024 – 2029